Arjun's Personal Design Theory

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Introduction: The Narrative of Interaction

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Design is seldom about pixels on a screen or plastic in a casing; it is about the quiet conversation between the intention of a designer and the reality of a user. As a graduate student at DePaul in Business Information Technology, my view of Human-Computer Interaction is informed by a singular confluence of high-pressure technical education and the cadences of everyday life. Whether I am exploring the architectural nuances of the AWS Management Console for my Cloud Practitioner certification or simply making sure my dog is safe on a Chicago sidewalk, I have come to understand that "good design" is a question of trust.

My design philosophy is based on three principles: Honesty, Autonomy, and Sustainability.

Honest design is transparent. As a result of my own need for order and attention to detail, I think a system should never obscure its status. Just as the Dyson vacuum's transparent container gives a "visual victory" over dust, a digital system such as Jira or AWS should give a clear, unobstructed view of the system status. When a design is honest, the user is in control. When it is opaque, it leads to "learned helplessness", a condition I know personally when the sheer complexity of cloud systems left me wondering if I was "made for this" industry.

Autonomy is the recognition of the user’s mental boundaries. As a cinephile, I am a believer in immersion. I think that an interface, whether it is an Amazon Fire TV or a cricket scorecard, should be a humble servant to the user’s ends, not a noisy trespasser. When an interface usurps my attention with auto-playing audio or busy advertisements, it oversteps the hallowed line of user autonomy. Design should enable the user to make educated decisions, such as understanding the price of a cloud offering or the status of a rental movie, before the moment of a click.

Finally, Sustainability explores the physical and mental stamina that our tools require. A tool that works flawlessly but leads to "trigger finger" fatigue or "complexity burnout" is a tool that lacks empathy. From the two-handed safety of a chainsaw to the "click" of a dog leash, our use should be sustainably ergonomic and mentally sustainable for the long term.

In the following Interaction Inventory, I examine these issues through eight artifacts that shape my world. Through the analysis of these common artifacts, I hope to transcend mere functionality and distill the principles that make technology not a burden to be managed, but an extension of our will.

Interaction Inventory

<u>My Digital Interfaces</u>

1. **AWS Management Console:** The AWS Console became a daily activity in my AWS Certified Cloud Practitioner study routine, where I utilized free credits to develop mini cloud projects. The global search bar at the top of the page is an essential, very visible affordance that enabled me to skip the very complex navigation menus to locate particular services. However, the system does not follow consistency heuristics, where different service interfaces (such as EC2 and S3) have completely different UI designs, requiring me to constantly relearn.

2. **Jira:** In my Scrum Master study routine, I often utilized Jira to practice sprint planning and learn the entire Agile process. The drag-and-drop feature of the Kanban board is a great example of the digital representation of post it notes, making the system state extremely intuitive. However, the page for creating new issues is often cluttered with required custom fields and small dropdown menus, significantly raising the cognitive load simply to create a new sprint issue.

3. **Cricbuzz Mobile App:** Cricket is a huge aspect of my culture in India, so I am extremely dependent on the Cricbuzz app to keep me updated during the ongoing T20 World Cup. The app is a great example of the "visibility of system status" heuristic, where the auto-refreshing ball-by-ball commentary allows me instant access to real-time information. Unfortunately, the main interface is greatly hindered by clutter, where aggressive banner ads are packed tightly between scorecards, often resulting in annoying accidental clicks.

4. **Amazon Fire TV Stick (Home Screen):** As a film buff who enjoys immersing myself in different worlds and experiencing the lives of characters on screen, the Fire TV home screen is my daily entry point into the world of cinema. The top navigation bar is an effective example of content chunking, where the main menu is divided into easily understandable categories (Home, Find, Live). Unfortunately, the hero banner at the top of the screen automatically plays video and audio previews when hovered over for even a second.

My Physical Artifacts

5. **Dyson V8 Absolute Vacuum:** As a confessed clean freak, I personally use my Dyson cordless vacuum cleaner every other day to maintain my space. The transparent dust container is an outstanding example of "visibility of system status," which enables me to immediately determine when the container is full without the need for abstract system indicators. Unfortunately, the continuous-hold power trigger necessitates constant physical pressure that severely contributes to hand fatigue when used extensively, disregarding the system's accessibility.

6. **RC Pets Extendable Dog Leash:** As my dog's safety is my primary concern, I personally use an extendable dog leash every day to shield her from larger dogs while allowing her to roam freely without straining her neck. The thumb-operated locking button is an outstanding example of an excellent immediate physical constraint that enables me to immediately stop the extension of the leash with one hand. Unfortunately, the flat ribbon-style cord tends to twist and get jammed inside the narrow housing, causing the system's fundamental functionality to completely fail during use.

7. **Homelite Chainsaw:** While I understand that the chainsaw does not meet the criteria of 'daily use' that the other artifacts do, the experience of using it in a high-stakes activity made the significance of design constraints impossible to ignore. I believed it was necessary to include it because it is the quintessential example of how forcing functions and feedback can literally be the difference between safety and disaster. After failing to meet the neighborhood deadline to place my Christmas tree outside for trash pickup, I was forced to take matters into my own hands and use a chainsaw to dispose of yard waste. The tool uses strong physical forcing functions; the main throttle physically cannot be activated unless a secondary safety switch is also being held. The feedback, however, is so aggressive that it completely overpowers any other system state, such as the chain tension release.

8. **CTA Ventra Card Reader:** I use the Ventra card readers as a student living in Chicago to ride the transit system. This device is a perfect example of multimodal feedback, as the quick and distinct "ping" sound, along with the bright green arrow, gives immediate feedback on a successful transaction in a loud and crowded space. However, the physical aspect of this device is not "discoverable" by touch. The flat, target-less surface is prone to "tap errors" where I have to hover my phone or card a few times to hit the perfect spot.

Personal Design Principals

1. The Autonomy Mandate

As I settle in to fully engage with a new world of cinema or watch the last few tense overs of a T20 World Cup match, my attention is a sacred thing. A system should never hijack my attention or my screen without my express mandate, but far too many systems aggressively prioritize engagement over my true intentions. The Amazon Fire TV grossly fails this mandate by loudly blasting audio and video previews before I have even made a conscious selection, while the Cricbuzz app crams jarring banner ads into the scorecard, leading to annoying accidental clicks. By doing so, these systems fail Jakob Nielsen's heuristic of User Control and Freedom, demonstrating that true usability is a matter of respecting the user's autonomy.

2. Fail-Safe Constraints

Crucial physical devices must always err on the side of absolute user safety rather than mere convenience. Whether I choose to take matters into my own hands after realizing the trash day has passed in our neighborhood or safeguard my dog from predators on our daily excursion, I must be able to rely on my devices to have physical constraints in place. The chainsaw does this flawlessly with an absolute forcing function, physically requiring a two-handed grip to turn the throttle on and thereby precluding disastrous accidental startup. In much the same way, my dog's retractable leash features a stiff thumb lock as a physical constraint, directly applying Don Norman's theory of Affordances and Constraints to provide a means for me to immediately stop the leash when danger is near.

3. Honest Visibility

Order and simplicity, both in my physical environment and in the smooth execution of an Agile sprint, depend on absolute visibility. I should never have to wonder whether a system is functioning or search through confusing menus to determine its status. My Dyson vacuum cleaner is a perfect example of a system that practices this heuristic flawlessly; the transparent plastic dust bin immediately conveys exactly when it is full, beautifully illustrating Jakob Nielsen's heuristic for Visibility of System Status. Unfortunately, although Jira's Kanban board is excellent for its visual mapping, its issue entry form conceals vital custom fields and log information behind layers of dropdown menus, making me search for the truth rather than being given an honest, upfront view of the sprint.

4. Spatial Predictability

Regardless of whether I am preparing for a Scrum Master certification or a Cloud Practitioner exam, my mind needs spatial predictability to work efficiently. When I interact with a system, I want the buttons and controls to remain in the same spot that I left them before. The Jira Kanban board is successful in this regard, as the columns are always in the same spot, no matter which project I am working on. But the AWS Management Console is always failing in this regard. When I move from the EC2 dashboard to S3, it is like moving to a different world altogether, where the UI patterns are all new, and I have to start learning all over again. This is a failure of Nielsen's heuristic of Consistency and Standards, and it proves that a lack of spatial predictability is a tax on the user's mental energy.

5. Ergonomic Sustainability

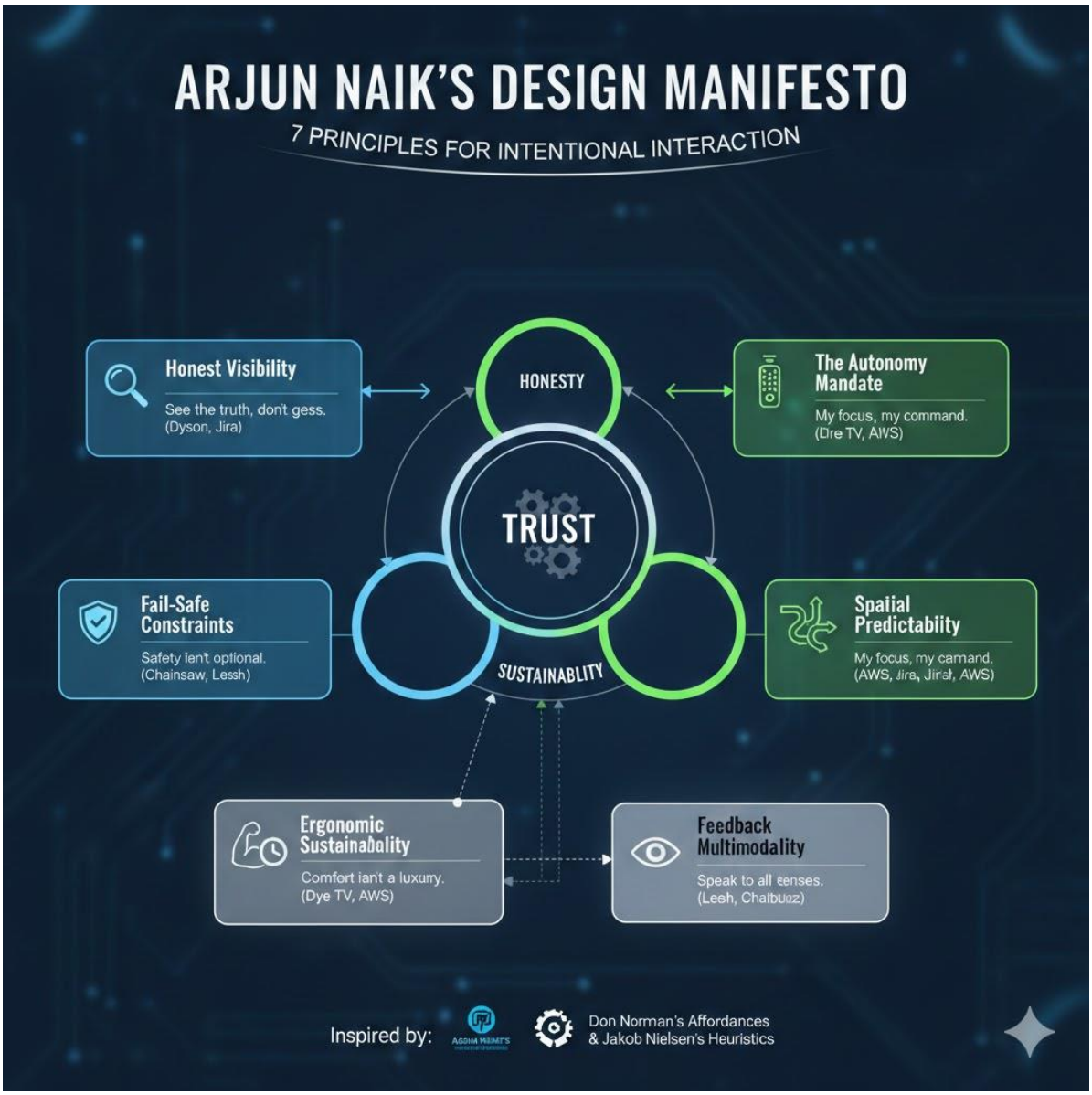
Design is not simply about the aesthetic or the functionality of an object in a vacuum; it is about how it feels in my hand over time. As a user of my Dyson vacuum cleaner every single day, I have come to realize that a "cool" feature can quickly become a physical burden if ergonomics are not considered. The constant-hold trigger on my Dyson vacuum cleaner requires constant pressure, which leads to hand fatigue, all in the name of conserving battery life rather than the comfort of the user. This is a lack of physical consideration that is also reflected in the digital world by the high information density of the AWS Console, which leads to a "visual fatigue" similar to the hand fatigue that I experience with my Dyson vacuum cleaner. A truly great design must be sustainable for the human body and mind over the entire duration of the task.

6. Feedback Multimodality

In the uncertain world of a daily walk, where I am always surveying for larger dogs to watch out for my own, a single feedback channel is often insufficient. I require feedback that my actions have been noted through multiple modalities so that I do not have to remove my eyes from my environment. My extendable dog leash is a perfect example of this; when I engage the thumb lock, I not only feel the button press, I hear a sharp, mechanical "click" that verifies the line is locked in place. This feedback is a perfect example of an immediate Visibility of System Status that allows me to respond immediately without looking away from my surroundings. I believe a similar, although louder, need exists for my chainsaw; the combination of the engine roar and the vibration informs me the saw is active, so I am aware of the system status through multiple senses before I make my first cut.

7. Mechanical Integrity

A system is only as robust as its weakest mechanical failure point. My dog's extendable leash is a lifesaver, but the flat ribbon cord is prone to twisting and jamming inside the housing, making the entire locking mechanism worthless during a walk. This is a failure of physical mapping; the housing inside doesn't take into account the three-dimensional, chaotic motion of a dog. I observe this same failure of mechanical integrity in digital systems, such as Cricbuzz, where the "retraction" of ads or videos is handled so badly that the interface literally stalls and jerks as I'm scrolling. If the underlying mechanics of a system can't take into account real-world motion, the user's faith in the entire system will be lost.



Case Study A: The Beloved Object – The Dyson Vacuum

"The Visual Victory of Cleanliness"

As a clean freak who appreciates meticulous order, the Dyson is not just a device it's a psychological feedback loop. My design aesthetic emphasizes the importance of Honest Transparency, and the Dyson satisfies this with its iconic transparent canister. In my personal experience, I don't have to rely on a sensor or a "bag full" indicator light; instead, I get an immediate visual reward for my efforts.

Why it Works (Principal Integration)

- Visibility of System Status (Nielsen): The transparency is what makes the "invisible" process of cleaning a "Visual Victory." For my OCD, being able to see the dust accumulation in the bin is what gives me a sense of closure and satisfaction that my space is truly safe and clean.
- Multimodal Feedback: When the vacuum hits a clog or a brush bar blockage, the motor pitch changes to a rhythmic pulsing. It's "talking" to me, telling me exactly where the system failure is without having to refer to a manual.

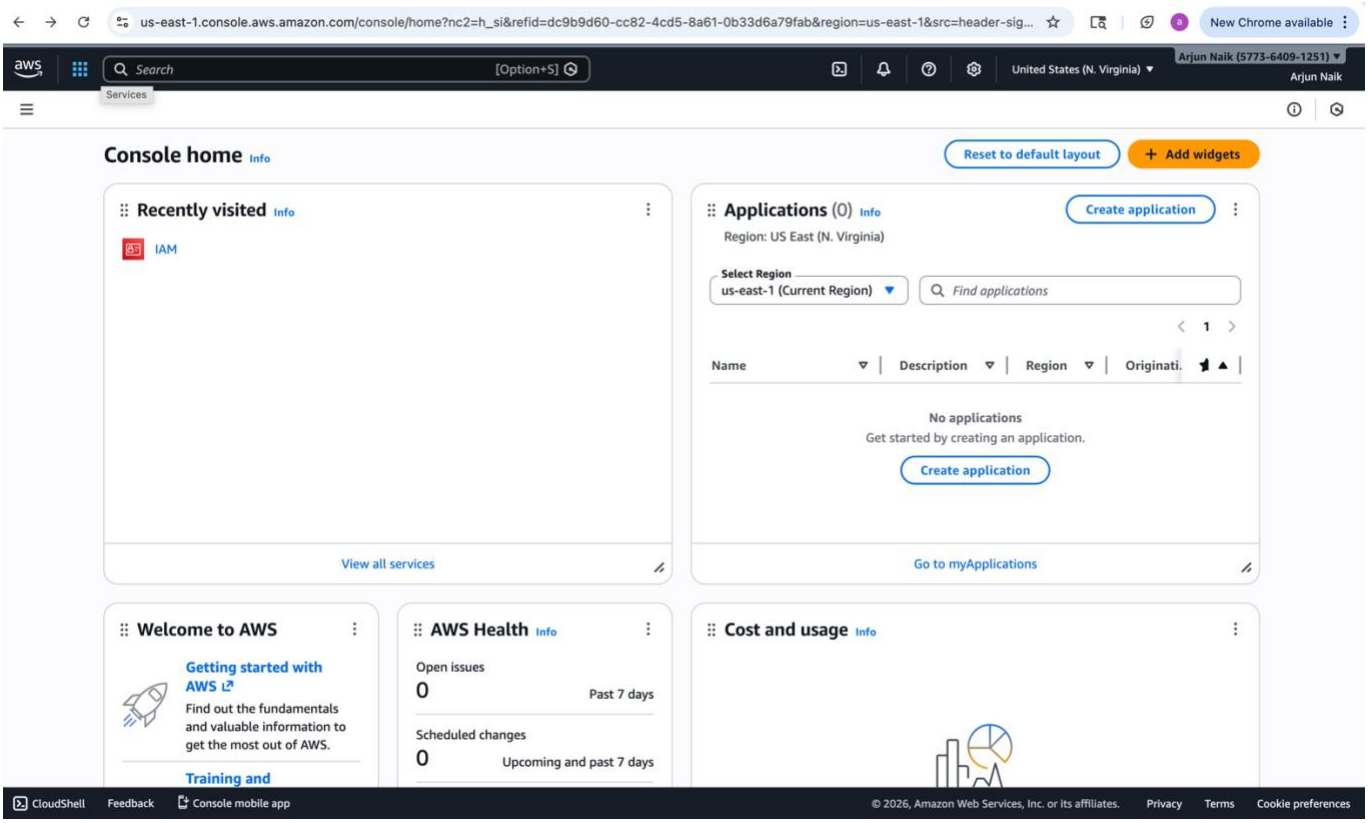
The One Flaw

Although it is a pleasure to use, it goes against my Ergonomic Sustainability principle. The "dead-man's switch" trigger design forces me to apply constant physical pressure. When I perform a deep clean, the physical toll on my hand starts to offset the psychological comfort of the clean floors. A "lock-on" toggle switch would turn this excellent tool into an ideal one.

Case Study B: The Problematic Design – AWS Management Console

"The Wall of Learned Helplessness"

During my prep work for the AWS CCP exam, I noticed a phenomenon in design that I'd like to refer to as Learned Helplessness. I would launch the console with the intention of creating something simple, only to be greeted by a "Complexity Wall." The overwhelming amount of technical vocabulary and the inconsistent UI between services such as EC2 and S3 left me with the impression that "I am not made for this," which is the definition of a failed user interface.



Screenshot of the AWS Management Console search bar and dashboard interface.

The Design Failures

- Violation of Consistency (Nielsen): Each service feels like it was designed by a different firm. This fails to provide the Spatial Predictability I require to feel comfortable.
- Information Overload: It disregards Progressive Disclosure. Rather than presenting me with the 20% of functionality I require as a beginner, it presents me with 100% of the enterprise-level complexity, immediately overwhelming my cognitive abilities.

My Proposed Solutions

- "Beginner Mode" Toggle: Just as I require my Firestick to display a clear "Rent/Buy" status before I click, AWS must have a "Learning Mode" that emphasizes Free-Tier eligible services and conceals enterprise-level configurations until I am prepared for them.
- Visual Cost Indicators: To alleviate the "fear of the bill," every action must have a clear, real-time estimate of its cost. If I could see exactly how many credits a service would consume before I launched it, the fear of "breaking the system" would dissipate, allowing for genuine exploration

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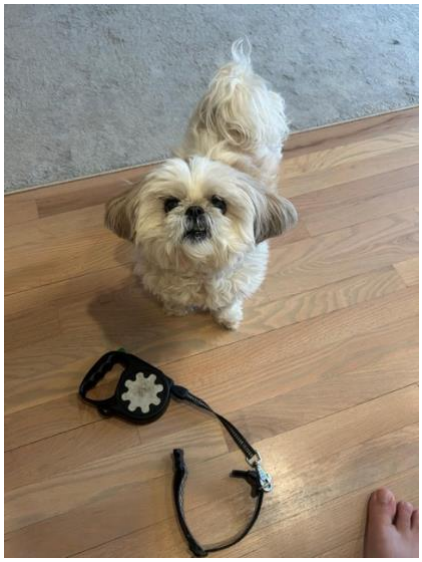
Conclusion: Beyond the Interface

This Interaction Inventory has completely changed the way in which I see the world around me. Prior to this exercise, I saw my Dyson, my dog's leash, and even the AWS Console as nothing more than tools-objects designed to complete a task so that I could get on with my life. But by taking the time to examine the "why" behind my successes and failures, I have come to understand that we are all engaged in a silent dialogue with the objects in our world.

I have come to understand that the "complexity wall" I experienced during my AWS preparation was not a reflection of my own intelligence but rather a gap in design empathy that could be bridged. I have come to understand that my need for order, which fuels my "clean freak" tendencies with my Dyson, is actually a design necessity for Honest Transparency that I must bring with me into my future career in IT management. Most importantly, I have come to understand that whether it is a high-stakes chainsaw or a simple cinephile's remote, the best designs are those that respect our time, our safety, and our humanity. As I move forward in my MS in Business Information Technology at DePaul, I no longer want to just build systems that work, I want to advocate for designs that make the user feel empowered, understood, and above all capable.

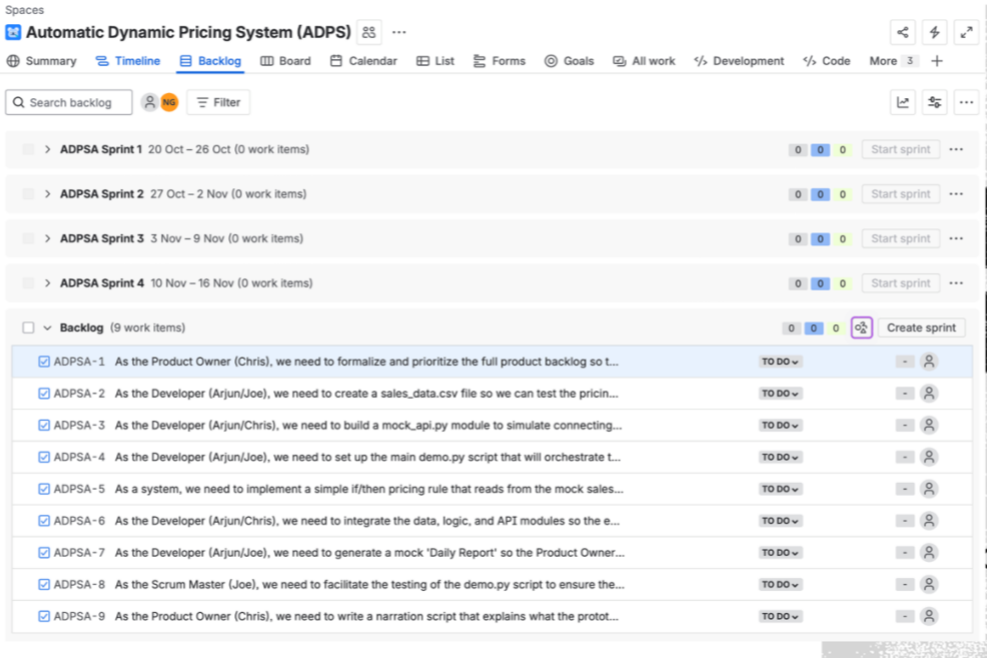
Appendix

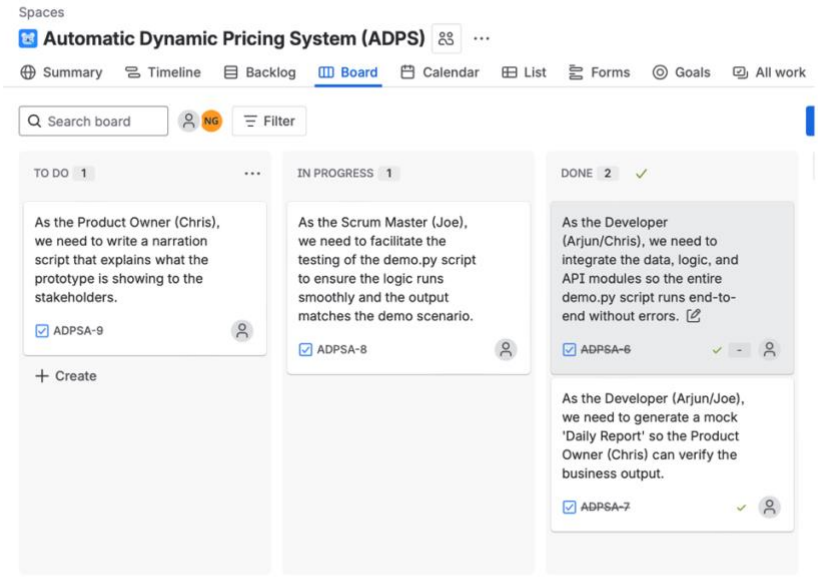
RC Pets Extendable Dog Leash Included to illustrate the physical constraints of high-stakes daily use. Although the thumb-lock offers excellent multimodal feedback (a tactile and audible "click") to guarantee the safety of my dog, the flat ribbon of the leash often jams—a harsh reminder that a lack of mechanical integrity can vitiate an otherwise good design in an instant



Chainsaw Included as an extreme example of fail-safe physical constraints. Operating this high-stakes tool requires an absolute forcing function, a dual-grip throttle that prevents catastrophic accidental activation. Combined with the intense multimodal feedback of its roar and vibration, it proves that dangerous tools must prioritize absolute user safety over simple convenience.

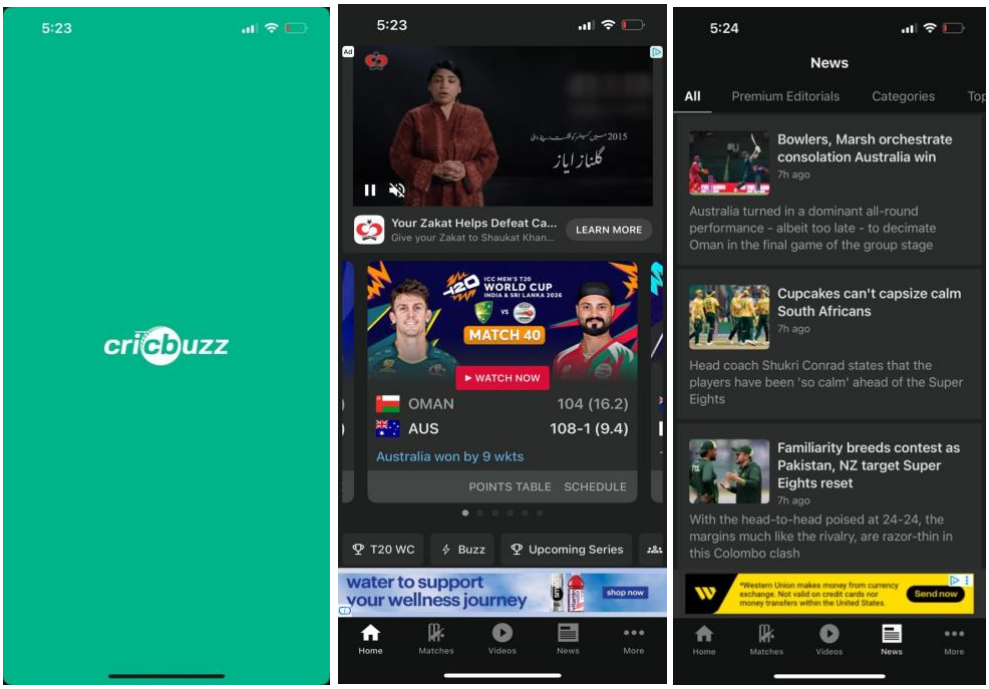
Jira Software Included to demonstrate the tension between spatial predictability and hidden complexity. While Jira's static Kanban columns expertly anchor my Agile workflows, its issue-creation screen buries critical custom fields behind nested dropdowns, a failure of honest visibility that forces unnecessary hunting for basic sprint information.



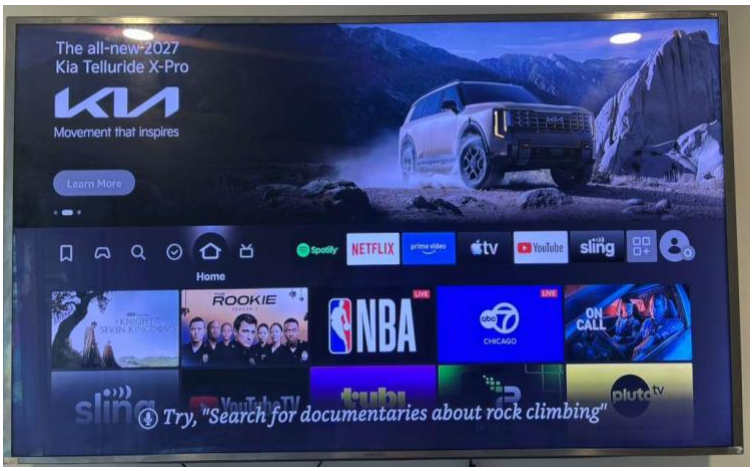


Cricbuzz Mobile App: Included to illustrate how aggressive monetization disrupts the user flow and violates my Autonomy Mandate. Following the staged interaction approach:

- **Stage 1 (Intent):** Opening the app for immediate, focused T20 live score updates.
- **Stage 2 (Interaction):** Scrolling through the dense data of the live scorecard.
- **Stage 3 (Disruption):** Dynamic banner ads load abruptly, causing the interface to jump. This mechanical stutter forces accidental clicks, completely hijacking user control.



Amazon Fire TV Stick: Included to illustrate a direct violation of my Autonomy Mandate by prioritizing aggressive system engagement over user intent. As a cinephile trying to browse deliberately, the interface constantly hijacks my focus with loud, auto-playing trailers before I even select. Furthermore, hiding critical "Rent/Buy" labels forces unnecessary clicks just to discover a movie's cost, completely derailing a seamless viewing experience.



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